PDS K-MEANS PRACTICUM - OPDRACHT 1: SERIËLE IMPLEMENTATIE BENCHMARK

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1. SYSTEM & BENCHMARK ENVIRONMENT

Cluster Node : VSC Compute Node (mindwell cluster)
Compiler : GCC 13.3.0 (g++)
Compiler Flags : -O3 -DNDEBUG -std=c++14
Debug Flags : None (Disabled for performance measurement)

2. TEST SETTINGS & INPUT PARAMETERS

Input File : datasets/100000x5.csv
K (Clusters) : 3
Repetitions : 20
Random Seed : 1337

3. EXECUTION TIME RESULTS

| Implementation | Best Dist Sq. | Time (seconds) |

| Custom Implementation (sequential) | 2.25117e+12 | 3.55336 |

| Reference Implementation (seq_icc) | 2.25117e+12 | 4.6005 |

```
sequential,1,1,datasets/100000x5.csv,1337,10,20,2.25117e+12,3.55336
sequential_icc,1,1,datasets/100000x5.csv,1337,10,20,2.25117e+12,4.6005
```

4. OUTPUT & CORRECTNESS

Verification script: compare.py

Output verificatie t.o.v. de reference implementation:

- Final Centroids (out.csv) : MATCH

- Cluster Traces (clustertrace.csv) : MATCH

- Centroid Traces (centroidtrace.csv): MATCH

```
19  # Output checks out: ['out.csv.1', 'out.csv.2']
20  # Output checks out: ['clustertrace.csv.1', 'clustertrace.csv.2']
21  # Output checks out: ['centroidtrace.csv.1', 'centroidtrace.csv.2']
22  # Done
```

- Time heeft wel een verschil: (4.6005 - 3.55336 = 1.04714)

⇒ onze implementatie is dus ongeveer 1 seconde sneller in deze run.